Course 4 · Week 3 — Bayesian, biomarkers, survival ML

Cheatsheet — biostats_courses

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Bayesian thinking

$$\underbrace{p(\theta \mid y)}_{\text{posterior}} \propto \underbrace{p(y \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

- Priors are part of the model — specify and defend them.
- Posterior summaries: mean, median, 95% credible interval.
- No p-values; use posterior probability of direction, Bayes factors, or LOO.

`brms` / Stan

```
library(brms)
fit <- brm(y ~ x + (1 | group), data = df,
           prior = c(prior(normal(0, 1), class = "b"),
                     prior(exponential(1), class = "sd")),
           chains = 4, iter = 2000, seed = 42)

summary(fit)
loo(fit)               # leave-one-out cross-validation
pp_check(fit)           # posterior predictive checks
```

Prior predictive check before fitting: simulate y from the prior and confirm the implications are plausible.

Biomarker statistics

Question	Statistic
Does a biomarker classify?	AUC, cut-point (Youden's index)
Does it add over an existing model?	ΔAUC, NRI, IDI, decision curves
Does it move prognosis?	Calibration plus discrimination
Is a cut-off reproducible?	Bootstrap CI on the cut-off

```
pROC::roc(y, biomarker) |>
  pROC::coords("best", best.method = "youden")
```

Survival ML

Model	R
Random survival forest	`randomForestSRC::rfsrc(Surv(...) ~ ., data)`
Gradient-boosted Cox	`xgboost` with `objective = "survival:cox"`
DeepSurv (conceptual)	`torch` with partial-likelihood loss

Evaluate with **time-dependent AUC**, **integrated Brier score**, and **IPA** (index of prediction accuracy = 1 − Brier(model) / Brier(null)). Account for censoring, for example with IPCW, and state the supported time horizon and censoring assumptions. Use a cohort-appropriate null risk.

```r
library(timeROC)
r <- timeROC(T = df$time, delta = df$event,
             marker = pred, cause = 1, times = c(365, 730))
```

External validation

- Never trust the apparent performance on training data.
- Minimum: split-sample or CV on internal data.
- Better: external validation in a second cohort.
- Report calibration slope / intercept, discrimination, and decision curves.

Decision rule for Week 3

- Sparse outcomes ⟶ assess separation and small-sample uncertainty; consider penalisation or a model with justified weakly informative priors.
- New biomarker ⟶ assess calibration, discrimination, and decision benefit on data independent of model development.
- Survival prediction ⟶ time-dependent Brier / IPA, not global AUC.
- Distinguish model development from external validation; evaluate transport to the intended setting before clinical adoption.

Common pitfalls

- Reporting a posterior with default flat priors on unscaled predictors.
- Selecting the Youden cut-off on the full data and using the same data to evaluate sensitivity / specificity.
- Quoting C-statistic at a single time point and calling it survival ML.
- Publishing a prediction model without TRIPOD-compliant reporting.

Further reading

- Gelman et al., *Bayesian Data Analysis*, 3e.
- Royston & Altman, *External validation of a Cox prognostic model.*